

# IEEE VR 2023 Workshops

## Workshop: ARES - Augmented Reality Enabling Superhuman Sports & Serious Games (2nd Annual Workshop)

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### ABSTRACT:

In the last years, the usage of Augmented Reality (AR) advanced into new frontiers through the widespread adoption of affordable mobile devices. For example, in Superhuman Sports, AR provides the potential to be a driving factor by revolutionizing team sports or by introducing novel game mechanics. In the field of Serious Games, AR can bring 3D serious content into the physical world to enhance immersion and knowledge transfer. In both areas, the full potential of this vision-based technology is still limited through the lack of framework support and tangible controller feedback for purely virtual objects.

We want to bring together experts from different computing and engineering disciplines to overcome the barriers of complex AR game concepts and to enhance interactions with augmented objects in a playful way in two important research fields:

The goal of Superhuman Sports is to surpass the limitations of the human body by incorporating technology. Cognitive and physical capabilities are augmented by involving rich interactions through tangible and virtual game objects. Concepts of team sports based on competition and the exploration of new experiences through novel senses and reflexes, are helping to augment traditional sports or to path the way towards new game ideas.

Serious Games combine the world of playing games for entertainment with the goal of learning in an almost subconsciously manner. Combining these two approaches requires meeting the demands of both. Learning content must be presented correctly and coherently, but on the other hand, it must be perfectly integrated into a game concept.

Today, there are still various unsolved and related challenges for AR sports concepts and AR Serious Games alike, which should be spontaneously, anywhere, playable on mobile/wearable devices.

Major challenges include, but are not limited to:

**AR Multiplayer Framework:** Reliably connecting mobile devices and supporting versatility in terms of augmentation platforms, is still an open issue. In a game engine such as Unity, there is up until this day a limitation in supporting self-made tracking algorithms, e.g., machine learning-based ones. On the other hand, game engines provide the important capability to efficiently render virtual content and provide the central game logic. We want to propose building blocks to solve such limitations and bring together mobile and microcontroller platforms into an AR framework.

**Tangible and Augmented Game Objects:** Traditionally, purely virtual game objects are used in AR games, because they can be easily manipulated in 3D space and hence require no tracking pipeline. On the other hand, they don't offer rich haptic feedback. To achieve realistic interactions, a combination of tangible and augmented game objects such as sports equipment or Serious Game objects are required. In this workshop we want to explore the realization of these rich tangible experiences.

**HMD Technology:** There is a vast gap in quality between cheap smartphone-based augmentation and sparsely available HMDs. During the workshop we want to emphasize development in the direction of novel display solutions and sensors to realize truly mobile AR experiences for dynamic game scenarios.

A white paper has been published as the basis for the workshop [1].

[1] Plecher, D., Eichhorn, C., & Klinker, G. (2022). ROAR-Role of Augmented Reality in Serious Games and Superhuman Sports, Technical Report (TUM-I2295), <https://mediatum.ub.tum.de/doc/1656728/1656728.pdf>

## **DEVELOPMENT OF A GAME SYSTEM TO PROVIDE ENJOYABLE PLAYFUL EXPERIENCES TO PEOPLE IN ALL STAGES OF DEMENTIA**

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In this work in progress paper [2], we discuss our work to develop an adaptive game system for people with dementia. The system makes use of projection-based AR and natural user interfaces to present game content in a manner which is engaging and allows for meaningful interaction. As the first step, we designed 7 different games, each based on a specific play experience. Later, we aim to investigate if each of these experiences could be enjoyed by users

with varying levels of cognitive impairments.

[2] Hu, W., Kuwahara, N., & Siriaraya, P.. (2023, March). Development of a Game System to Provide Enjoyable Playful Experiences to People in All Stages of Dementia. In 2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). IEEE.

## **THE OPPORTUNITY TO COLLECT KINETIC FORCES DURING A ROUND OF GOLF USING WEARABLE DEVICES: SINGLE-SUBJECT CASE STUDY – THE FULL SWINGS**

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This single-subject case study [3] compared a golfer's swing kinematics and kinetics on a driving range and golf course using a pair of pressure sensor socks and a markerless motion capture solution. Results indicated a significant difference in the right and left plantar pressure percentages in wedge swings and 7-iron chest sway ( $p < 0.05$ ). Findings suggest that wearable and mobile technology can measure discrete and longitudinal training and

gameplay to enhance a golfer's performance.

[3] Luczak, A. L. et al. (2023, March). The Opportunity to Collect Kinetic Forces during a Round of Golf using Wearable Devices: Single-Subject Case Study – The Full Swings. In 2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). IEEE.

## **BONKERS BORDERS - A SERIOUS AR GAME OF BOUNDARIES UNFITTING AND CONSEQUENCES UNWITTING**

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Bonkers Borders [4] is a serious Augmented Reality (AR) game that depicts a territorial struggle between rival imperial factions to lay claim to a newly discovered fictional continent. The project aims to be a satirical simulation of the detached imperialist approaches when carving up land borders on maps. The goal is to create a people-centric territory control game inspired by real-world events that could also be used to broaden the fields of serious and hybrid games, by providing an engaging and immersive experience with pedagogical, ludological, and technological relevance. The project takes the form of a hybrid physical-digital game that allows players to shape the game world through actions and objectively view the consequences of their decisions through a companion

app. The implementation of this hybrid game was evaluated by players, with findings indicating significant interest in the potential for technology in tabletop games. The key aspects of such an application include computation, hidden information management, and immersion in the form of AR to realize a stationary 2D world in 3D. These findings also bode well for the nascent field of hybrid games.

[4] Rangarajan, S. et al. (2023, March). Bonkers Borders - A Serious AR Game of Boundaries Unfitting and Consequences Unwitting. In 2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). IEEE.

## **DEVELOPING A CYCLIST 3D GAMEOBJECT FOR A MIXED REALITY INTERACTION FRAMEWORK**

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Mixed Reality could be used as a safe testbed to capture behavioural data of pedestrians when interacting with other road users, such as cyclists, by introducing a virtual cyclist. However, for such real to virtual interaction to be more realistic, the cyclist GameObjects in Unity have to look and move similar to a real cyclist. The existing available 3D cyclist are currently not well suited for such interacting scenarios as they are assets based on prerecorded animations. In this work-in-progress paper [5] we discuss our design of a 3D cyclist GameObject for interactions in Unity. We also explain how we

exploit a mixed reality visualisation tool (i.e., Microsoft Spectator View) for our setup, allowing a second person view of a Hololens MR application. We further discuss the scenarios that we consider for a pedestrian-cyclist interaction with this work.

[5] Kamalasanan, V. et al. (2023, March). Developing a Cyclist 3D GameObject for a Mixed Reality Interaction Framework. In 2023 IEEE Conference on Virtual Reality and 3D User Interfaces (VR). IEEE.